

VisionGroup Symphony Assessment Task

JSON API for contact lens option selection

Design and implement a JSON API that handles product configuration constraints for contact lenses. Not all parameter combinations are valid - your API must dynamically filter available options based on user selections to prevent invalid combinations.

Background

Contact lens parameters (power, base curve, diameter, etc.) must be selected sequentially, but certain combinations aren't manufacturable. Your API needs to intelligently filter remaining options based on current selections, working in any parameter order while enforcing business rules.

Mock this example

Example general

Parameter 1 options: A,B,C

Parameter 2 options: X,Y,Z

Where A/Y and C/Z are not allowed.

The selection process always selects one parameter after another. This will allow us to filter out not allowed combinations.

Example combinations

1. No parameter selection

GET /parameter

```
JSON
{
  parameter1: [A,B,C],
  parameter2: [X,Y,Z]
}
```

2. First parameter: parameter 1 B. Options for second parameter 2: X,Y,Z

GET /parameter?parameter1=B

```
JSON
{
  parameter1: [B],
  parameter2: [X,Y,Z]
}
```

3. First parameter: parameter 1 A. Options for second parameter 2: X,Z

GET /parameter?parameter1=A

```
JSON
{
  parameter1: [A],
  parameter2: [X,Z]
}
```

4. First parameter: parameter 2: Z. Options for second parameter 1: A,B

GET /parameter?parameter2=Z

JSON

```
{  
  parameter1: [A,B],  
  parameter2: [Z]  
}
```

Example reality

Here is a real life example of this problem from our competition

<https://discountlens.ch/p/AO1DMT-30/acuvue-oasys-max-1-day-for-astigmatism-30>

discountlens by Urbach Optik

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Home > Contact lenses > Toric Lenses > ACUVUE OASYS MAX 1-Day for Astigmatism 30

ACUVUE OASYS MAX 1-Day for Astigmatism 30

Box of 30 | Box of 90

SAVINGS CHF 3.70

☒ Left ☐ Right

Sphere * Spheres

Cylinder * Cylinders

Axis Axis

Radius 8.5

Diameter 14.5

10° 20° 30° 40° 50° 60° 70° 80° 90° 100° 110° 120° 130° 140° 150° 160° 170° 180°

CHF 40.90

ADD TO CART

Shipping information

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ACUVUE OASYS MAX 1-Day for Astigmatism 30

Box of 30 | Box of 90

SAVINGS CHF 3.70

☒ Left ☐ Right

Sphere * +3.50 Spheres

Cylinder * Cylinders

Axis Axis

Radius 8.5

Diameter 14.5

10° 20° 30° 40° 50° 60° 70° 80° 90° 100° 110° 120° 130° 140° 150° 160° 170° 180°

Please insert your values

Price CHF 40.90 Prices include VAT

CHF 40.90 Per unit

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Shipping information

Task

Create an API to retrieve available parameters while respecting the not allowed combinations.

1. Think about how to solve this problem and write down at least 2 possible solutions
2. Discuss advantages and disadvantages of your chosen solution over the not chosen solution from 1
3. Create your own sample data and a test api that implements this behavior. Create an endpoint to query and demonstrate
4. Create a public github repo and send the repo link to (ulrich.heck@visiongroup.io and saulius.cepauskas@visiongroup.io)
5. Present and discuss your solution to VisionGroup